Lecture 3

The exercises in the previous lecture were very easy with solutions $\begin{pmatrix} 77 & 7 \\ 12 & 27 \end{pmatrix}$ for first question and *Yes*, with $\lambda = \mu = 0$ for the second question. Please also do more exercises on your course notes, and should you not be sure about your work. Take a screenshot and send an email for feedback.

In the previous lecture notes, we have shown that the matrix multiplication is associative, satisfies the left and right distributive laws (in relation to matrix addition) but fails to be commutative. What about the story of matrix multiplicative identities and matrix multiplicative inverses? we shall postpone this to later lectures after we have also revisited solving systems of linear equations.

Matrix transpose

Definition 6. *Let $A = (a_{ij})$ be an $m \times n$ matrix, then the matrix transpose $A^T = (b_{ij})$ is an $n \times m$ matrix defined by making each row of A the corresponding column of A^T, that is, we have $b_{ij} = a_{ji}$ for each $1 \le i \le m$ and $1 \le j \le n$.*

Examples Below is a list of three matrices on the left and their corresponding matrix transpose on the right (in respective order).

$$\begin{pmatrix} 0 & 0 & e \\ 2 & 7 & 6 \end{pmatrix}, \begin{pmatrix} 0 & 8 \\ 2 & 7 \end{pmatrix}, \begin{pmatrix} 7 \\ 2 \\ 0 \\ 2023 \end{pmatrix} \qquad \begin{pmatrix} 0 & 2 \\ 0 & 7 \\ e & 6 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 8 & 7 \end{pmatrix}, \begin{pmatrix} 7 & 2 & 0 & 2023 \end{pmatrix}$$

The matrix transpose operation has the following properties.

Proposition 5. *Let $A = (a_{ij})$ and $B = (b_{ij})$ be $m \times p$ matrices, $C = (c_{ij})$ a $p \times n$ matrix and λ a number. Then we have*

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(\lambda A)^T = \lambda A^T$
- $(BC)^T = C^T B^T$

The first three properties of the above proposition are very easy to verify, please do so, and the last property is in your tutorial 01. You will have to check that the matrix expression on the left hand side is of the same size as the one on the right hand side and that their corresponding entries are equal.

We shall later also be interested in matrices A such that $A^T = A$, let $A = (a_{ij})$ be an $m \times n$ matrix for the equality to hold we must have $m = n$ (to ensure same size requirement) and $a_{ji} = a_{ij}$ for all $1 \le i \le m$ and $1 \le j \le n$ (corresponding entry equality requirement).

Definition 7. *Let $A = (a_{ij})$ be an $n \times n$ matrix, then A is called symmetric if $a_{ji} = a_{ij}$ for all $1 \le i \le n$ and $1 \le j \le n$.*

Examples Matrices $\begin{pmatrix} 0 & 2 \\ 2 & 7 \end{pmatrix}$, $\begin{pmatrix} 0 & 2 & 3 \\ 2 & 7 & 6 \\ 3 & 6 & 0 \end{pmatrix}$ are symmetric and matrix $\begin{pmatrix} 0 & 2 & 0 \\ 2 & 7 & 6 \\ 3 & 6 & 0 \end{pmatrix}$ is not symmetric.

Note that symmetric matrices as per our definition are symmetric about the diagonal.

Given an $m \times n$ matrix A, we also get two symmetric matrices, that is, $m \times m$ matrix AA^T and $n \times n$ matrix A^TA. Indeed, let us consider matrix AA^T, we have that $(AA^T)^T = (A^T)^T A^T = AA^T$ by using matrix transpose properties and so matrix AA^T is symmetric. Verify similary, that matrix A^TA is also symmetric.